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Problem I. Dumb Problem II

Input file:	standard input
Output file:	standard output
Time limit:	1.5 seconds
Memory limit:	1024 megabytes

For a permutation p of length n, we define $f(p)$ to represent the position of the prefix maximum. Specifically, $f(p) = \{i | 1 \le i \le n, \text{s.t. } \forall 1 \le j < i, p_i > p_j\}$.

Similarly, let $g(p)$ represent the set of numbers of the prefix maximum, that is, $g(p) = \{p_i | i \in f(p)\}$.

Now there are k independently and uniformly randomly chosen permutations $p_1, p_2, \ldots, p_k$ from all permutations of length n (can be repeated). Little A wants to know how many essentially different $g(p)$ there are expected among these permutations. The goal is to find out how many essentially different sets are expected among $g(p_1), g(p_2), \ldots, g(p_k)$? The answer should be output modulo 998244353.

Input

A single line containing two integers n, k ($1 \le n, k \le 5000$).

Output

A single line containing an integer representing the result of the answer modulo 998244353.

Examples

standard input	standard output
1 1	1
3 2	388206139
3 4	408232647
112 646	928854225

Note

For the first sample, the answer is 1.

For the second sample, the answer is $\frac{31}{18}$.

For the third sample, the answer is $\frac{1711}{648}$.